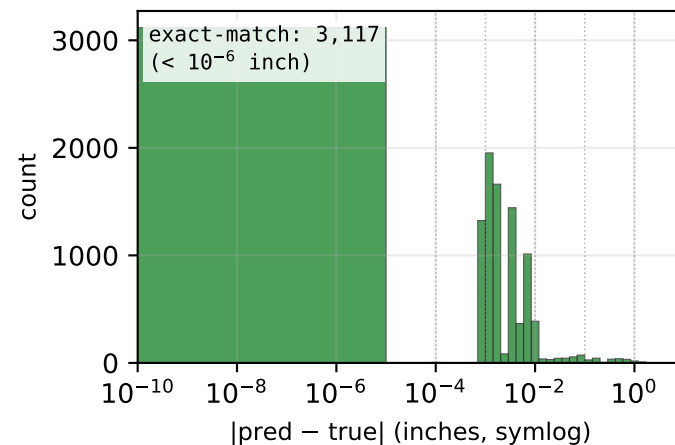
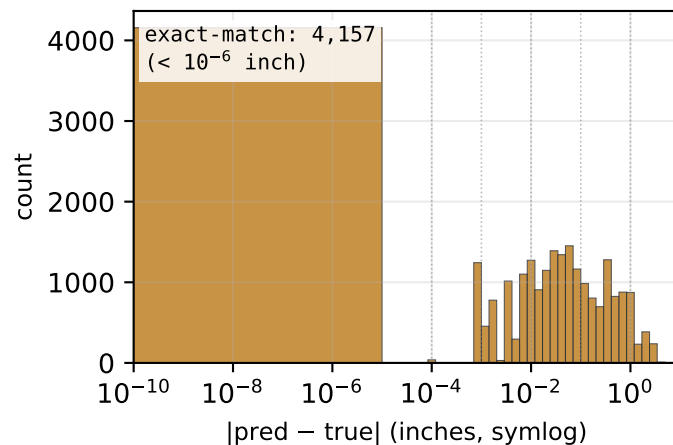
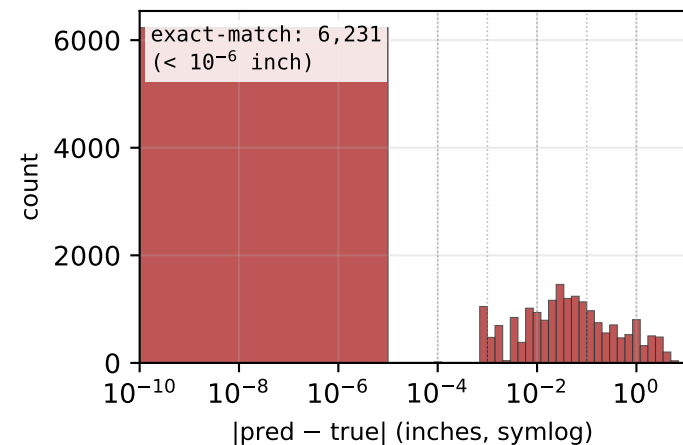
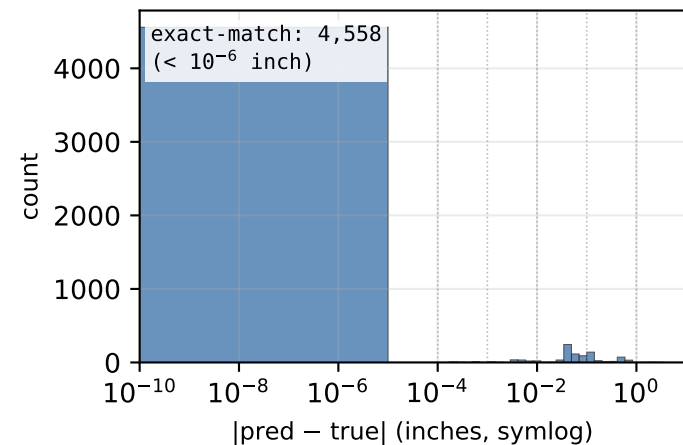


Per-axis prediction-error magnitude (V8 5-fold test, subsampled to 5,000 positions per axis where available)

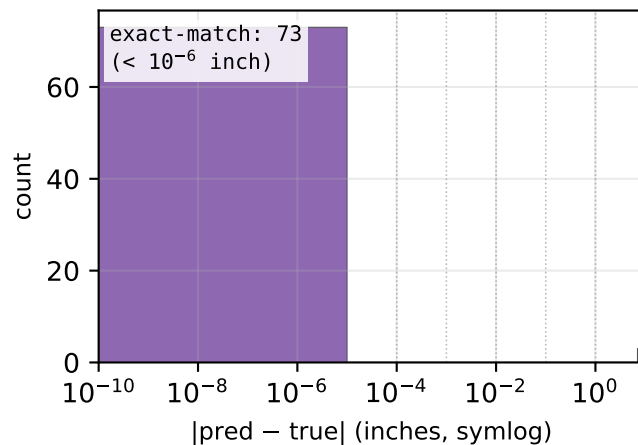
X: n=25,000 exact=25% MAE=0.251 P95=1.671 Y: n=25,000 exact=17% MAE=0.199 P95=0.999 Z: n=11,853 exact=26% MAE=0.012 P95=0.010



R: n=5,486 exact=83% MAE=0.023 P95=0.101



F: n=82 exact=89% MAE=1.309 P95=12.919



Reading guide:

Errors concentrate at exact-match (left) if the model recovers the precise value.

Bimodal distribution (mass at exact-match AND a far mass) is a mode-collapse signature.

Vertical reference lines mark the precision tiers: 10^{-4} inch (ten-thousandths), 10^{-3} (thousandths), 10^{-2} (hundredths), 10^{-1} (tenths), and 1 inch.